

Project Assignment 2024-2025

Building a Domain-Specific Search Engine

Objectives:

- The design and implementation of a document retrieval system that can index and search effectively text documents in a collection.
- The development of information retrieval algorithms and the evaluation of their performance using common evaluation metrics such as accuracy, recall and F1 score.
- To provide a user-friendly interface for users to enter queries and retrieve relevant documents.
- The practical experience in various information retrieval techniques, such as Boolean retrieval, Vector Space Model and Probabilistic retrieval models.

Project Structure and Phases:

1. Domain Selection and Project Planning (Week 1)

Objective: Kick off the project by choosing a domain and planning the development phases.

Activities:

- Research and select a domain that is interesting and has accessible data.
- Draft a project proposal, including the domain, types of documents, and a brief plan on how you intend to collect, preprocess, and index the data.

Milestone: A 1-page project proposal outlining domain selection, motivation, and expected challenges.

2. Data Collection and Preprocessing (Week 2-3)

Objective: Collect relevant data and preprocess it for indexing.

Activities:

- Collect domain-specific documents (e.g., scraping websites, using open datasets, or APIs like PubMed).
- Preprocess the data (tokenization, stopword removal, stemming/lemmatization).
- Store the processed documents in a suitable format for indexing (e.g., JSON, CSV).

Milestone: A report on data collection and preprocessing, along with the cleaned dataset.

3. Building the Inverted Index (Week 4)

Objective: Construct an inverted index for efficient retrieval of documents.

Activities:

- Understand how inverted indexes work (terms, postings lists).
- Implement an inverted index using Python or Java.
- Store the index for later use in query processing.

Milestone: A working inverted index with a description of its design and implementation.

4. Query Processing (Week 5-6)

Objective: Implement query processing, allowing users to search the indexed documents.

Activities:

- Implement basic query operations such as Boolean search (AND, OR, NOT).
- Parse and process user queries into a format that can be searched against the index.
- Begin handling ranking based on basic term frequency (TF-IDF) calculations.

Milestone: A functioning query processor capable of handling Boolean queries.

5. Ranking and Relevance Feedback (Week 7-8)

Objective: Enhance the search engine with document ranking based on query relevance.

Activities:

- Implement ranking of search results using TF-IDF or BM25.
- Optionally implement a simple relevance feedback system to improve results based on user input.

Milestone: An improved search engine with ranked results based on user queries.

6. Evaluation and Performance Metrics (Week 9-10)

Objective: Evaluate the performance of the search engine using standard IR metrics.

Activities:

- Introduce evaluation metrics: precision, recall, F1-score, and Mean Average Precision (MAP).
- Create a set of test queries to evaluate system performance.
- Report on system performance, identifying strengths and weaknesses.

Milestone: An evaluation report with detailed analysis of the system's performance using appropriate IR metrics.

7. User Interface Development (Week 10)

Objective: Design and implement a simple user interface for the search engine.

Activities:

- Build a web-based or command-line interface where users can input queries and view results.
- Ensure the interface is intuitive and easy to use.

Milestone: A basic user interface with functionality to submit queries and display ranked results.

8. Final Report and Submission (Week 11-12)

Objective: complete the project and submit all final materials.

Activities:

- Prepare a final report that details the system's design, challenges faced, implementation, and evaluation results with discussion. In the report present the search engine with screenshots, demonstrating its functionality and explaining the design choices.

Deliverables:

- **Final Report:** Comprehensive project report in pdf format. In the first page of your report under your name, write the link of your code (see below)
- **Code Submission:** Well-documented codebase of the entire search engine in Jupyter Notebook file (ipynb) uploaded in repository (colab or Kaggle or GitHub). The Notebook should

include your code, results and comments. Provide the link of the code in the first page of your report under your name.

Assessment criteria:

- **Code and Functionality (50%):** Correctness and efficiency of the search engine's indexing, querying, and ranking components.
- **Reports and Documentation (35%):** Quality of final report, and documentation of the code.
- **Evaluation and Analysis (15%):** Depth of evaluation using relevant metrics and discussion of results.

Due date: 13/01/2025 (late submissions are not accepted)